

12 (long)

(a) What is the magnitude and direction of the electric field at a position 5 m from a single point charge Q in air? [5]

(b) What is the magnitude and direction of the electric field at a position (O) that has 5 point charges of Q arranged at angles $\theta = 0^\circ, \pm 60^\circ$ and $\pm 90^\circ$ with $R = 5$ m in air as shown in Fig. 10a? [10]

(c) A line of charges with positive charge density $\rho = 5 \mu\text{C m}^{-1}$ is placed in air. It has a shape as shown by Fig. 10b, where the semicircle has a radius of $R = 5$ m and the two straight sections each have a length of $L = 5$ m. What is the magnitude and direction of the total electric field at the centre of the semicircle (O)? [15]

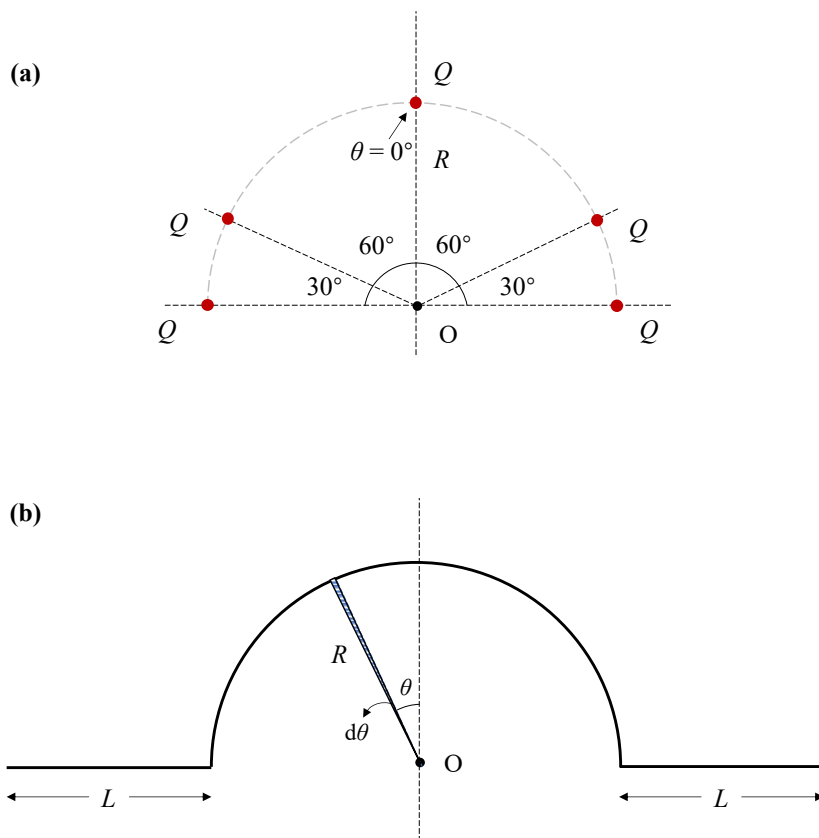


Fig. 10

END OF PAPER

THIS PAGE IS BLANK